

IP-SAKTI SAHAYAK

TKDL-Only Family Dataset — Lamiaceae / Tulsi

Purpose: Create a family-wise knowledge base for the proposed feature: when a plant/formulation has relevant prior-art or patent concerns, identify other TKDL-listed plants in the same botanical family as *potential alternatives for further research*. This document does not establish patent freedom or non-infringement.

Source rule: plant names and TKDL evidence in this document are restricted to the official TKDL public material identified below. No external plant database is used as a source of plant entries.

1. Anchor plant

Tulsi — Ocimum sanctum Linn.

TKDL also records the name as Tulsi / sacred or holy basil in its plant-name index.

2. Related TKDL plant candidates

TKDL-listed related plant	Relation to anchor	Similarity signal for prototype	Use in model
Ocimum basilicum Linn.	Same genus / family	High botanical proximity; compare traditional use/formulation evidence	Candidate
Ocimum canum Sims	Same genus / family	High botanical proximity; compare evidence before ranking	Candidate
Ocimum gratissimum Linn.	Same genus / family	High botanical proximity; compare evidence before ranking	Candidate
Ocimum basilicum var. purpurascens Benth.	Same genus / family	Very close taxonomic proximity; treat as variant-level candidate	Candidate
Ocimum album Linn.	Same genus / family	Very close taxonomic proximity; compare evidence before ranking	Candidate
Mentha arvensis	Same family	Lower proximity than same-genus Ocimum; compare formulation/use overlap	Candidate
Mentha piperita	Same family	Lower proximity than same-genus Ocimum; compare formulation/use overlap	Candidate

3. Similarity methodology for IP-SAKTI

Do not use family membership alone. For each candidate, calculate a research similarity score from: (1) same genus/family, (2) same plant part where documented, (3) overlap in traditional indication, (4) overlap in formulation role/ingredient usage, and (5) strength of TKDL evidence. The model should rank candidates as High / Medium / Low similarity and explain the evidence.

Recommended output: “Potential alternative candidate — requires botanical, formulation, efficacy and patent/legal verification.” Never output “legally safe” solely because the candidate is in the same family.

4. TKDL evidence relevant to Tulsi

Official public TKDL plant-name index: <https://www.tkdil.res.in/tkdil/langdefault/common/Utility/Keyword/Plant-Name.asp>

The TKDL index explicitly lists Ocimum basilicum, Ocimum basilicum var. purpurascens, Ocimum canum, and other Ocimum entries. The Unani/common TKDL plant index also lists Ocimum basilicum, Ocimum canum, Ocimum gratissimum and Ocimum sanctum.

Public examination record 1111/CHE/2010: TKDL material cites Ocimum sanctum (Tulasi) among plants used in traditional formulations/uses; the record includes a formulation with Sacred/Holy basil as a single ingredient for general tonic and another formulation containing Ocimum sanctum for an immuno-suppressive/emaciating disease. Official record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1111_CHE_2010.asp

Public examination record 1506/MUM/2011: TKDL references include Ocimum sanctum (basil) and describe its traditional use for jaundice; the examination record also raises Section 3(p) and inventive-step issues. Official record: https://www.tkdil.res.in/tkdil/langdefault/common/Examinarreport/reports/1506_MUM_2011.asp

Public examination record 7795/DELNP/2008: the record lists Ocimum sanctum together with other ingredients in a formulation and states that the claimed medicinal properties were known in traditional knowledge. Official record:

5. Dataset fields to create

family | **anchor_plant** | **candidate_plant** | botanical_name | TKDL_name/synonyms | plant_part | traditional_use | formulation | formulation_role | source_text | TKDL_public_case_id | official_TKDL_url | similarity_score | similarity_reasons | patent/IP_flag | verification_status

Important: The uploaded project reference also instructs the AI not to conclude that an ingredient is “unpatentable” merely because it appears in a TKDL-related case, and to distinguish Section 3(p), novelty and inventive step.

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6. Verification status

The public TKDL plant index is suitable for confirming that a botanical name is represented in TKDL. A plant-name listing alone does **not** prove that a particular plant has the same formulation, therapeutic use, or patent relevance as the anchor. Those fields must be populated only when supported by a specific TKDL record/source.

This is Family 1 of the planned 10-family dataset. The same schema should be reused for the remaining families.